Practical Assignment: Data Foundations for Machine Learning

Field	Information
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Topic: Local Market Food Prices Dataset

1. Introduction

Data Science is the process of collecting, analyzing, and extracting useful insights from data using statistics, programming, and domain knowledge. Machine Learning is a subset of Artificial Intelligence that learns patterns from data to make predictions or decisions.

For this assignment, I created and documented a dataset containing food product prices collected from local markets. The purpose of the dataset is to understand how different factors such as product type, weight, market location, day of observation, and import status influence product prices.

This project demonstrates the fundamental concepts learned in Lessons 1–3, including data collection, features and labels, supervised learning, dataset quality, and the Data Science lifecycle.

2. Dataset Collection Method

The dataset was created through manual observation and recording of food product prices from different local markets.

For each observation, information about the product, weight, market location, day of collection, import status, and selling price was recorded.

The final dataset contains:

- 83 samples (rows)
- 5 features (input variables)
- 1 label (output variable)

The dataset intentionally contains some real-world data quality issues such as missing values, duplicate records, and inconsistent text entries to provide opportunities for preprocessing and cleaning.

3. Features and Label

In Machine Learning, features (X) are the input variables used for prediction, while the label (y) is the output variable that the model attempts to predict.

Features (X)

Feature	Description
Product	**Type of food product**
Weight_kg	**Weight of product in kilograms**
Market	**Market where the product was observed**
Day	**Day of data collection**
Imported	**Whether the product is imported (Yes/No)**

Label (y)

Label	Description
Price_USD	**Product selling price in US Dollars**

4. Dataset Structure

The dataset contains:

- Rows (Samples): 83
- Features: 5
- Label: 1
- Total Columns: 6

Sample Dataset

Product	Weight_kg	Market	Day	Imported	Price_USD
Rice	2	HamarWeyne	Tue	No	3.08
Pasta	5	Hamar Weyne	Wed	Yes	6.47
Cooking Oil	10	Suuqa Xoolaha	Sat	No	20.43
Beans	3	Hamar Weyne	Tue	No	2.86
Rice	1	Hamar Weyne	Sat	Yes	1.85

Each row represents one product observation, while the entire table forms the dataset.

5. Quality Issues in the Dataset

Real-world data is rarely perfect. Several quality issues exist in this dataset and will need preprocessing before Machine Learning can be applied.

Missing Values

Some records have missing values in columns such as:

- Product
- Weight_kg
- Market
- Day
- Price_USD

These missing values may need to be removed or replaced using appropriate imputation techniques.

Typographical Errors

Some text values are inconsistent.

Examples:

- Hamar Weyne

- HamarWeyne
- Bakara
- Bakaraa

These entries refer to the same locations but are written differently.

Duplicate Records

The dataset contains duplicate rows that may distort analysis and should be identified and removed.

Inconsistent Categories

Some imported values appear as:

- Yes
- No
- Y
- N

These categories should be standardized during preprocessing.

6. Learning Type

This dataset represents a **Supervised Learning** problem.

According to Lesson 2, supervised learning uses labeled data where the correct output is known.

In this dataset:

Features (X):

- Product
- Weight_kg
- Market
- Day
- Imported

Label (y):

- Price_USD

Since the target variable (Price_USD) is available for each observation, the model can learn the relationship between product characteristics and price.

7. Machine Learning Use Case

The dataset is suitable for a **Regression** problem because the target variable is a continuous numerical value.

The trained model could be used to estimate product prices based on product and market information.

Potential applications include:

- Market price prediction
- Business planning
- Inventory management
- Economic monitoring

8. Relationship to the Data Science Lifecycle

This dataset follows the Data Science lifecycle discussed in Lesson 1.

Problem Definition

Determine whether product prices can be predicted using product and market characteristics.

Data Collection

Food prices were collected from local markets and recorded in a structured dataset.

Data Cleaning

Missing values, duplicate records, and inconsistent categories must be corrected.

Exploratory Data Analysis (EDA)

Visualizations and summary statistics can be used to understand pricing patterns.

Modeling

A regression model can be trained using the collected data.

Evaluation

The model's performance can be measured using metrics such as Mean Absolute Error (MAE) and Mean Squared Error (MSE).

Communication

The final results can be presented using reports, tables, and visualizations.